DATA STRUCTURES LAB SE COMP

Group B Practical no 3

Experiment 3: A call center receives incoming calls, and each call is assigned a unique customer ID. The calls are answered in the order they are received. Your task is to simulate the call queue of a call center using a queue data structure.

- **addCall(customerID, callTime):** Add a call to the queue with the customer ID and the call time (in minutes).
- **answerCall():** Answer and remove the first call from the queue.
- **viewQueue():** View all calls currently in the queue without removing them.
- **isQueueEmpty():** Check if the queue is empty.

```
from collections import deque


# Create queue
call_queue = deque()


def addCall():
    customerID = input("Enter Customer ID: ")
    callTime = int(input("Enter Call Time (minutes): "))

    call_queue.append((customerID, callTime))
    print("Call added successfully.")


def answerCall():
    if not call_queue:
        print("Queue is empty. No call to answer.")
    else:
        customerID, callTime = call_queue.popleft()
```

```
        print("Answered Call")
        print("Customer ID:", customerID)
        print("Call Time:", callTime, "minutes")

def viewQueue():
    if not call_queue:
        print("Queue is empty.")
    else:
        print("\nCurrent Call Queue:")
        for customerID, callTime in call_queue:
            print("Customer ID:", customerID,
                  "| Call Time:", callTime, "minutes")

def isQueueEmpty():
    if not call_queue:
        print("Queue is empty.")
    else:
        print("Queue is not empty.")



# Menu
while True:
    print("\n----- CALL CENTER QUEUE -----")
    print("1. Add Call")
    print("2. Answer Call")
```

```
    print("3. View Queue")
    print("4. Check Queue Empty")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        addCall()

    elif choice == 2:
        answerCall()

    elif choice == 3:
        viewQueue()

    elif choice == 4:
        isQueueEmpty()

    elif choice == 5:
        print("Program terminated.")
        break

    else:
        print("Invalid choice.")
```

Output

---- CALL CENTER QUEUE -----

1. Add Call

2. Answer Call

3. View Queue

4. Check Queue Empty

5. Exit

Enter your choice: 1

Enter Customer ID: C101

Enter Call Time (minutes): 5

Call added successfully.

Enter your choice: 1

Enter Customer ID: C102

Enter Call Time (minutes): 8

Call added successfully.

Enter your choice: 3

Current Call Queue:

Customer ID: C101 | Call Time: 5 minutes

Customer ID: C102 | Call Time: 8 minutes

Enter your choice: 2

Answered Call

Customer ID: C101

Call Time: 5 minutes